2020 COMP3222/9222 Practical Exam

Instructions

- **Please ensure that you upload files, where required, as you progress through the exam – submissions will not be permitted after the time allowed for the exam has expired.**
- **Please note that only your last submission for each part will be marked.**

How will the exam be conducted?

- The Final Practical exam will be held on the **afternoon of Monday 30 November.**
- The exam will be held from **4:00 pm until 6:20 pm** on **WebCMS**.
- There are **4 questions** in total.
- The exam is worth **30 marks** in total. The exam contributes 25% to your final assessment in the course. You must score at least 40% of the available marks in this exam in order to pass the course.
- The exam has a **140 minute time limit** (consideration for 10 minutes reading time and time to upload files with momentary glitches in connectivity are included in this time allowance).
- All times stated for the exam are in Sydney time (AEDT = UTC+11).

What exam materials can be used?

- Make sure you have a quiet location with good internet connectivity where you can work on the exam undisturbed for the full 140 minutes.
- No exam materials are provided. You will need paper and pens and a means of scanning and uploading answers to some of the questions.
- You may use **any offline resources**, including your course notes, personal notes, textbooks, and lab files.
- You may not request help from any person or service apart from the Course Coordinator, who will be available by phone (+61 2 9385 7384), email (o.diessel@unsw.edu.au) and on Teams for the duration of the exam.

No help is allowed during the exam

- **You must not communicate with anyone while you are taking the exam.**
- **You are not permitted to use any online resources, e.g. searching for answers to questions using the Internet.**
- **Failure to comply with the above, if detected, will result in disciplinary action.**
- If you experience difficulties during the exam, or have a question about the exam, contact the Course Coordinator by phone (+61 2 9385 7384), email (o.diessel@unsw.edu.au) or via Teams as soon as possible.

Check-list of things to prepare when sitting an online exam

https://student.unsw.edu.au/sites/all/files/uploads/group271/UNSW%20Online%20Exam%20Preparation%20Checklist.pdf

Technical issues

If you experience a technical issue before or during the exam, you should follow these instructions:

- Take screenshots of as many of the following as possible:
  - error messages
  - screen not loading
  - timestamped speed tests
  - power outage maps
  - messages or information from your internet provider regarding the issues experienced
- All screenshots must include the date and time the issue occurred.
- Contact the Course Coordinator by phone (+61 2 9385 7384), email (o.diessel@unsw.edu.au) or via Teams immediately and advise of the issue.
- Submit a Special Consideration application immediately at the conclusion of your assessment and upload your screenshots.

Academic misconduct & Plagiarism

- **You must treat this exam as you would treat an in-person, supervised and invigilated exam**. Any evidence of academic misconduct will be treated seriously.

Student Declaration

You will complete a digital declaration prior to commencing the exam. By completing the declaration you are declaring that:

- As with any task conducted through UNSW, I am aware that online assignments and examinations require me to comply with the UNSW Student Code of Conduct (https://student.unsw.edu.au/conduct).
- I have read and understood the University requirements in respect of student academic misconduct outlined in the Student Code of Conduct (https://student.unsw.edu.au/conduct) and Annexure 1 of the Student Misconduct Procedures (https://www.gs.unsw.edu.au/policy/documents/studentmisconductprocedures.pdf)
- It represents my own work and that I did not give or receive assistance in any way that would violate the UNSW Student Code of Conduct (https://student.unsw.edu.au/conduct),
- I have taken proper and reasonable care to prevent this work from being copied by another student
- ***I understand the following actions, among others, constitute Academic Misconduct:***
  - ***Allowing others to complete an online task in my place, either in whole or in part.***
  - ***Taking or sharing screenshots of the assessment task or my responses unless expressly permitted by UNSW.***
  - ***Otherwise duplicating or distributing the assessment task in whole or in part.***
  - ***Communicating any assessment content, question or topic to any person prior, during or following the assessment unless expressly permitted by UNSW.***

- ***Providing or seeking assistance from unauthorised information sources including, but not limited to, <u>the Internet</u>, any programs, chat servers, web services, or contact with any external parties to communicate, collude or get assistance in answering examination questions either prior to or during an online exam session.***
- I understand any breach of the above terms and conditions may result in university disciplinary action.

Special consideration

- If you cannot take your exam for reasons beyond your control, you may be eligible to apply for special consideration. Please see the special consideration information on the webpage https://student.unsw.edu.au/special-consideration
- Note that the fit-to-sit rule applies as usual. If you start the exam, you are deeming yourself able to complete the exam. If during the exam you feel unwell to the point that you cannot continue with the exam, you should take the following steps:
  - Stop working on the exam and take note of the time
  - Make contact with the Course Coordinator by email (o.diessel@unsw.edu.au) or via Teams to advise that you are unwell
  - Immediately submit a Special Consideration application saying that you felt ill during the exam and were unable to continue
  - If you were able to advise your Course Coordinator of your illness during the assessment, attach screenshots of your attempt to do so to your Special Consideration application

Problem Specification

Many physical phenomena need to be counted; for example, the number of collisions of a specific type in a particle collider, or the number of photons above a given energy in a telescope image. Often these samples need to be collected and categorized in volumes and at speeds that render processor-based solutions infeasible.

In this problem, you are required to design a simple logic-based sample counter. You are required to design a circuit that counts the number of samples in a given input stream that are above a given threshold.

A diagram of the circuit ports, indicating the required connection to both the DE0 and DE1 board resources is illustrated in Figure 1.

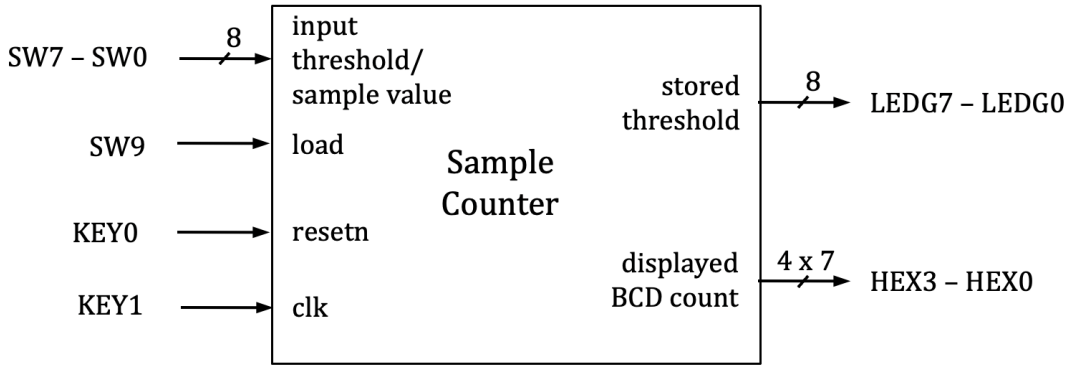

Figure1. Sample counting circuit.

The circuit uses 8 input switches, a load switch, an active-low resetn button, a push button switch to manually simulate the clock input, 8 green LEDs to display the stored threshold value in binary and the four 7-segment displays HEX3-HEX0 to report the current count value in decimal.

The circuit operates as follows:

1. The circuit is asynchronously reset ***at any time*** by asserting the active-low resetn signal (by depressing button KEY0). This causes the threshold and count values to be initialized to 0.

2. After the circuit has been reset, an unsigned 8-bit threshold value is set by the user via switches SW7 – SW0. This value is loaded into the circuit by asserting the load signal (SW9) and waiting for a rising clock edge (generated by releasing KEY1). The stored threshold value is output via LEDG7 – LEDG0. The load signal must be deasserted before the circuit can commence counting samples (steps 3 & 4).

3. The circuit is now ready to count sample values that exceed the threshold. An 8-bit unsigned sample value is presented to the circuit via switches SW7 – SW0. The value is loaded into the circuit by asserting the load signal and waiting for a rising clock edge.

4. Upon *the first rising clock edge after load is deasserted again*, the loaded sample value is compared with the stored threshold and the count is incremented if the sample value is strictly **greater than** the threshold value. The count value is displayed as a decimal number on HEX3 – HEX0.

5. The circuit continues to count samples by repeating steps 3 & 4 until the resetn signal is asserted.

Notes:

1. Recall that the push buttons are pulled low, i.e., provide a 0 input when they are pressed. Therefore, a rising edge is generated when the button is released.

2. The load signal can remain asserted for any number of clock pulses during steps 2 & 3. The value that is stored before proceeding to the next step is the value set when the last rising clock edge occurred while the load signal was asserted.

3. The count value should be reported in decimal using the four hexadecimal displays. It is therefore recommended that you use a 4-digit BCD counter within the Sample Counter circuit to store the count value. The counter digits should each be converted to 7-segment display driver signals for output to HEX3 – HEX0.

4. The circuit does not check for overflow of the count value. The count value will therefore cycle back to 0 when it counts the 10,000th sample exceeding the threshold.

A simulation of the sample counter in operation is illustrated in Figure 2. In this simulation, the input value is named *t_s_val*, the name of the stored threshold value is *stored_t* and the name of the driver signals for the least significant 7-segment display of the count value is *disp_count0*. Note that *disp_count0* displays the signals for the 7-segment display rather than displaying the decimal digits 0, 1, 2…, but these appear as annotations below the waveform.

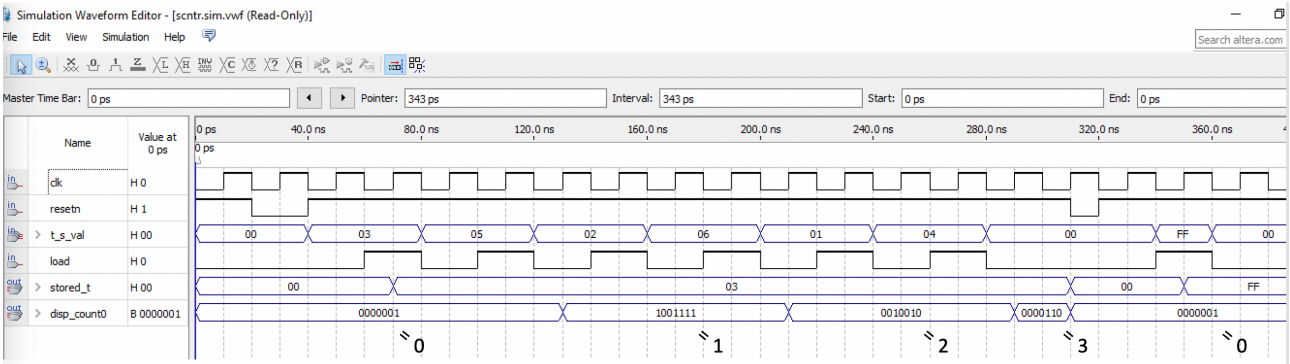

Figure 2. Simulation of the sample counter circuit.

You are provided with a Quartus Project Archive with pin assignments that map the circuit to the board you are using. The archives are named `DEn-scntr.qar`, whereby you should use the archive named for the board you are using **n = 0 for DE0 and n = 1 for DE1**. The archive contains the outline of the top-level design entity for this problem, called `scntr`. (**To facilitate testing, please do not modify the top-level ENTITY description for `scntr`.**) The archive contains the waveform file used to generate the timing diagram above. The archive also includes a bitstream for the working circuit, named `scntr-soln.sof`, in the `output_files` subdirectory. This bitstream can be used to program your board by "adding the file" to the list of programs for the device you are using.

Please ensure you download and use the correct archive for the board you are using!

Submission

Please upload your files regularly.
<u>This will avoid delays when students submit simultaneously at the end of the exam.</u>
Please note that only your last submission will be marked.
YOU MUST SUBMIT YOUR WORK WITHIN THE TIME ALLOWED.
Late submissions WILL NOT be accepted.

You are required to submit:

1. [5 marks]

   A neat & legible listing of the **pseudocode** for this problem and a neat & legible sketch of your **initial ASM chart** for your design. Please take care with Mealy & Moore type outputs. Upload a SINGLE PDF, JPG or PNG format file containing your answer using the WebCMS submission tab for Part 1. of the exam.

2. [4 marks]

   A neat & legible sketch of the **datapath** for your design. The sketch <u>MUST indicate the name of all signals as used in your VHDL description</u> and correspond with the names used in your refined ASM chart of 3. Your datapath sketch <u>MUST indicate the number of bits used for each signal and the size of all components.</u> Upload a SINGLE PDF, JPG or PNG format file containing your answer using the WebCMS submission tab for Part 2. of the exam.

3. [3 marks]

   A neat & legible sketch of a **refined copy of your ASM chart** from 1. This diagram <u>MUST use the same signal names and state names as used in your VHDL description and MUST indicate the state transitions and outputs of the control path</u>. Upload a SINGLE PDF, JPG or PNG format file containing your answer using the WebCMS submission tab for Part 3. of the exam.

4. [18 marks]

   A **Quartus Project Archive** of your completed design. <u>To facilitate testing, do not modify the top-level ENTITY description for `scntr`.</u> Upload a SINGLE QAR format file containing your answer using the WebCMS submission tab for Part 4. of the exam.

   Your solution will be marked according to the following scheme:

   a. [15 marks]

      Still testing and debugging VHDL code – we will assess the completeness and correctness of your description.

   b. [2 marks]

      Simulation complete – make sure we have your waveform file in the directory as we will check your simulation progress.

c. [1 marks]

Implementation complete – we will test your design on the board.